

CrewSight

Phased rollout & governance plan · 2026–27 season and beyond

Prepared by Altitude HD · 14 June 2026 · aligned with AHD LiveStream regatta calendar

Suggested path: Free proof-of-concept at NZ Master Rowing (16 units) → KRI purchases **50 devices at its own risk** (no grant) for early-season regattas → ~50 additional units for the second half, with **grant funding sought for the cumulative 100-device programme** pending Phase 1 success → **+170 devices** in 2027–28 (100 already in operation) for full **270-device** fleet. Timeline, costs, funding model, and governance below.

2026–27 regatta calendar

Source: Altitude HD LiveStream hub calendar (ahd-livestream.vercel.app).

Dates	Event	Phase	Fleet target
19–20 Sep 2026	New Zealand Master Rowing	PoC — free trial	16 (AHD loan)
14–15 Nov 2026	KRI Memorial Rowing Regatta	Phase 1 — first purchase	50
28–29 Nov 2026	KRI Club Regatta	Phase 1 — first purchase	50
11–13 Dec 2026	KRI Christmas Regatta	Phase 1 — first purchase	50
29–31 Jan 2027	North Island Rowing Champs	Phase 2 — mid-season expansion	~100
26–28 Feb 2027	North Island Secondary School Rowing Regatta	Phase 2 — mid-season expansion	~100
15–20 Mar 2027	Maadi Cup	Phase 2 — mid-season expansion	~100
2027–28 season	Full national programme (Karāpiro + Ruataniwha + partners)	Phase 3 — +170 units (270 total)	270

Rollout timeline

Aug–Sep 2026 — PoC preparation

Altitude HD provisions 16 GPS handsets, CrewSight app, cloud hosting, and IoT connectivity. On-water validation, charging setup trial, ops monitor training with KRI volunteers.

19–20 Sep 2026 — NZ Master Rowing (PoC)

No charge to Karāpiro Rowing. Demonstrate fleet maps, zone alerts, hull events, and live-stream integration. Debrief within 2 weeks; go/no-go on Phase 1 purchase.

Oct 2026 — Phase 1 decision & procurement

KRI commits to 50 handsets (+ optional mounting kits) **without grant funding — KRI bears the risk**. Invoice 1 (CAPEX handsets) + Invoice 2 (season platform ± data). Industrial charging rack installed at number hut. Device IDs and APN checks completed before Memorial. Begin grant application scoping for Phase 2 (contingent on PoC/Phase 1 outcomes).

Nov–Dec 2026 — Phase 1 operations (50 devices)

KRI Memorial, Club Regatta, Christmas Regatta. KRI owns charging, daily prep, and on-site ops. Altitude HD provides platform support during events and ingest monitoring.

Jan 2027 — Phase 2 expansion decision

Subject to Phase 1 success, procure ~50 additional handsets ahead of North Island Champs. **Grant funding target:** support the cumulative **100-device** programme for the balance of 2026–27 (Phase 2 hardware + connectivity; platform already paid in Phase 1). SIMs activated for balance of season (~2 months). Cumulative fleet ~100 for NICC, NISS, and Maadi block. If grant is not secured, KRI board decides whether to self-fund Phase 2 or defer.

Mar 2027 — Season review

Review device condition, data usage, charging workflow, and partner hire model. Plan 2027–28 full 270 fleet and annual service agreement.

2027–28 season — Phase 3 expansion (+170 units)

Purchase **170 additional handsets** (100 already owned and operational from 2026–27) to reach 270 total. Deploy across Karāpiro programme and partner events (incl. Lake Ruataniwha via hire model). Second charging cabinet / mobile case; spares pool; annual CrewSight service fee (see below). Grant or RNZ capital partnerships may be revisited for national scale.

Funding model

Phase	Devices	Funding approach	Rationale
PoC	16	Altitude HD — no charge to KRI	Demonstrate value before any KRI capital commitment
Phase 1	50	KRI self-funded — no grant	KRI accepts risk on initial fleet; proves ops model at Memorial, Club, and Christmas regattas
Phase 2	+50 (100 total)	Grant funding sought (pending Phase 1 success)	Applications target the 100-device season programme for 2026–27 — handsets, connectivity, and deployment at NICC → Maadi; platform fee already covered in Phase 1
Phase 3	+170 (270 total)	KRI / RNZ / partners — TBC	Incremental CAPEX only; 100 units already in operation from prior season

Suggested grant narrative (Phase 2): After a successful PoC and Phase 1 deployment, KRI applies for community or sport development funding (e.g. Rowing NZ regional development, Sport NZ community resilience, local trust, or equipment grants) to complete a **100-device CrewSight fleet** for the remainder of the 2026–27 national regatta block. Indicative grant scope: Phase 2 add-on of \$7,000.00 ex GST (data included) or the cumulative 100-device season total of \$20,500.00 ex GST if funders prefer a single programme figure. Grant applications should reference Phase 1 operational evidence (uptime, charging workflow, ops monitor usage, and event debrief outcomes).

Grant funding is not guaranteed. Phase 2 proceeds only on KRI board approval if funding falls short. Phase 1 commitment is independent of grant outcomes.

Indicative costs (ex GST)

Handsets \$130.00/unit · Platform \$6,000.00/season (flat) · IoT data \$5.00/SIM/month · Optional mounting \$50.00/boat

Stage	Scope	Data included	Data excluded	Notes
PoC	16 units · NZ Master Rowing	\$0.00	\$0.00	AHD investment (~\$1,697.44 internal cost ex GST)
Phase 1	50 handsets · 4-month season platform & SIM	\$13,500.00	\$12,500.00	KRI self-funded · decision after PoC · Nov–Dec regattas
Phase 2 add-on	+50 handsets · ~2 mo SIM (no second platform fee)	\$7,000.00	\$6,500.00	Grant target (pending Phase 1) · Jan 2027 · NICC → Maadi
2026–27 cumulative	100 handsets · one platform fee	\$20,500.00	\$19,000.00	Incl GST: \$23,575.00 · grant scope reference

Stage	Scope	Data included	Data excluded	Notes
Phase 3 (2027–28)	+170 handsets · 4-mo SIM (new units) · platform · 270 total fleet	\$31,500.00	\$28,100.00	Incremental only — 100 units already owned · incl GST: \$36,225.00

Phase 3 costs cover **170 additional units** only (\$22,100.00 handsets + \$3,400.00 SIM + \$6,000.00 platform). Full greenfield 270-device season reference: \$46,500.00 ex GST (see CrewSight-Season-Proposal-270-devices.pdf). Existing 100 handsets may require SIM renewal in 2027–28 if data-included — budget separately if applicable.

Optional mounting (per phase)

Fleet size	Mounting (\$50.00/boat)
50 devices (Phase 1)	\$2,500.00 ex · \$2,875.00 incl
100 devices (cumulative)	\$5,000.00 ex · \$5,750.00 incl
Phase 3 add-on (+170)	\$8,500.00 ex · \$9,775.00 incl
270 devices (full fleet)	\$13,500.00 ex · \$15,525.00 incl

Field battery life

Supplied handsets (One NZ Smart **M26**) have a **5,000 mAh** battery. **Field test 14 Jun 2026:** M26 at ~1 Hz GPS drained ~3.1 %/h (96% → 76% over 6.5 h) — estimated **~32 h** from full charge at continuous high-rate tracking. An 8-hour active regatta day at 1 Hz uses ~25% battery. Standard mixed regatta profile (1 s / 5 s / 30 s intervals) uses less — validated against Samsung Galaxy S21 reference in the same vehicle test (M26 median GPS 3.3 m vs S21 5.2 m).

Scenario	Implication
Single regatta day (≤8 hr on water)	No mid-day charging required — ~25% at 1 Hz; less on mixed profile
Multi-day regatta block	Overnight industrial-cabinet charge between days; spot-check battery % in ops monitor before dispatch
Long sessions / warm-up days	~32 hr headroom at continuous 1 Hz; mixed profile extends further
Handset settings	Battery optimisation unrestricted · Location “Allow all the time” · Notifications enabled

Prior lab reference (Samsung A1 @ 30 s GPS): ~3.7 %/h (~27 h from full charge). Actual runtime varies with cellular coverage, screen-off discipline, ambient temperature, and battery age. Replace or rotate units that no longer hold charge through a full regatta day during annual health checks.

Partner hire — South Island & Beach Sprints events

For Lake Ruataniwha (Maadi, South Island Rowing) and Beach Sprints NZ coastal events, KRI retains device ownership. Partner organisations pay a **device hire fee to Karāpiro Rowing** (separate from CrewSight platform fees and Altitude HD operational/broadcast charges). Indicative hire: cost-recovery plus margin — to be agreed per event (e.g. per-device-day or flat event block). Altitude HD supports ingest and maps; event owner retains data (see below).

Governance & frequently asked questions

Who owns, operates, and maintains the devices and SIMs?

Karāpiro Rowing Inc (KRI) — devices are purchased by KRI as CAPEX. KRI operates day-to-day: charging, pre-regatta checks, distribution to boats, and return/storage. SIMs are either managed by KRI under the data-excluded option, or billed through Altitude HD / One NZ under the data-included option with KRI responsible for physical handset care. Altitude HD maintains the CrewSight platform, cloud hosting, and app updates.

How are devices charged at scale?

Recommend an **industrial USB charging cabinet** (multi-port, keyed, ventilated) with **bespoke racking mounted on the number hut wall** at Lake Karāpiro. Labelled slots per device ID; overnight charge between regatta days; spot-check battery health in the ops monitor before dispatch. With ~32 hr measured headroom at 1 Hz (5,000 mAh M26, 14 Jun 2026 field test), a full overnight charge covers a standard 8-hour regatta day without mid-day top-up. Phase 3 (270 fleet) may add a second cabinet or mobile charging case for away events.

How do South Island Rowing and Beach Sprints NZ contribute for their events?

Device hire fee payable to Karāpiro Rowing, on top of any operational or broadcast fees. KRI ships a subset of the fleet (or hires additional units) to the event; hire covers wear, logistics, and KRI staff time. CrewSight platform access for that event is included in the season platform fee or pro-rated event add-on as agreed. Beach Sprints coastal events may require waterproof mounting — optional \$50.00/boat kits.

Who owns the data?

The event owner — typically the regatta organising committee or national body for that event. GPS tracks, alerts, and session history are exported or archived per event. KRI as device owner does not automatically own competitor location data; usage follows event privacy policy and Rowing NZ guidance.

Who pays for live-stream value?

Rowing NZ — proposition to Milford (and existing AHD broadcast partners). Live maps, vMix overlays, and enhanced storytelling are part of Altitude HD’s broadcast offering, funded through Rowing NZ’s community / development budget or sponsor packages — not through KRI’s handset CAPEX. CrewSight feeds the broadcast; it is complementary to the safety/ops use case.

Who owns the intellectual property?

Altitude HD — CrewSight software, cloud architecture, overlay designs, and integration IP remain Altitude HD property. KRI owns physical handsets and mounting hardware. Event data ownership is separate (above).

Who owns future value propositions?

Rowing NZ and Altitude HD — national features (analytics, sponsor integrations, NZ-wide dashboards, Maadi broadcast enhancements) are for RNZ and AHD to develop and commercialise. Karāpiro Rowing benefits as the anchor deployment and device custodian but is not the commercialisation partner for national propositions.

Can Karāpiro Rowing use these devices for other events?

Yes. KRI-owned handsets may be used for club regattas, training camps, and approved third-party hires provided CrewSight platform terms are met (device registration, event naming, data handling). Hire to external events is a KRI revenue opportunity; national calendar events should prioritise RNZ/KRI agreed schedule.

Five-year improvement roadmap

Horizon	Focus
2026–27	PoC → 100 devices · charging rack · ops monitor · zone alerts · vMix overlay · partner hire model pilot at Maadi
2027–28	270 fleet · spares pool (≈5%) · automated pre-regatta health checks · Postgres purge policy · Ruataniwha & beach sprint playbooks
2028–29	Predictive maintenance alerts · improved hull-event ML · multi-venue ops dashboard · RNZ national analytics layer

Horizon	Focus
2029–30	Device refresh cycle begins · optional LTE fallback · integration with RowIT progression and results APIs
2030–31	Full fleet refresh option · 5G-ready handsets · sustained annual service model · exportable event data packages for RNZ

Ongoing maintenance, replacement & future fees

Device maintenance (KRI + AHD)

- **KRI:** daily charge checks, physical cleaning, mounting inspection, secure storage, lost-device process
- **Altitude HD:** app updates, cloud uptime, ingest monitoring, SIM provisioning (data-included), remote diagnostics
- **Annual health check:** test GPS, battery, and upload path for each handset before season start; replace failed units from spares pool

Replacement costs

Handsets: budget **\$130.00** per replacement unit (current sell price; cost \$86.09 ex GST). Recommend a **5% spares pool** at full scale (≈ 14 units at 270 fleet) — $\sim \$1,820.00$ ex GST initial spares budget. Mounting kits: \$50.00/replacement. Assume **20–25% fleet refresh every 3–4 years** (battery and ruggedness).

Future annual service fee (indicative)

From 2027–28 onward, once the full fleet is deployed, suggest a flat **\$10,000.00/season** CrewSight platform & support fee (ex GST, \$11,500.00 incl GST) covering cloud hosting, app updates, event support, and ingest monitoring — in lieu of scaling platform fee with device count. IoT data and handset CAPEX remain separate line items. Per-regatta support can be quoted ad hoc for away venues if beyond season scope.

Next steps: (1) Confirm PoC at NZ Master Rowing · (2) KRI board decision on 50-unit self-funded purchase by Oct 2026 · (3) Draft Phase 2 grant application (100-device programme) contingent on Phase 1 success · (4) Rowing NZ / Milford discussion on broadcast funding · (5) Charging rack specification with KRI facilities · (6) Partner hire terms draft for Maadi / Beach Sprints · (7) Review legal section below with respective counsel.

Legal

This section summarises legal and compliance matters for discussion between Karāpiro Rowing, Rowing NZ, event partners, and Altitude HD. It is **not legal advice**. Formal terms will be set out in separate written agreements (platform licence, supply/purchase, hire, and data-processing arrangements as applicable).

Platform disclaimer. CrewSight is a situational-awareness and fleet-visibility platform for rowing operations and broadcast. It is **not certified rescue or safety-of-life equipment** and does not replace official race rules, patrol boats, qualified officials, emergency services, or personal flotation requirements. Alerts and maps are aids to decision-making only; operators remain responsible for on-water safety at all times.

1. Status of this document

This rollout plan is an indicative commercial and operational proposal. Costs, timelines, and governance answers are subject to board approval, partner agreement, and execution of binding contracts. Nothing in this document creates a binding obligation until signed by the relevant parties.

2. PoC (NZ Master Rowing) — free trial

- Altitude HD provides up to 16 handsets, connectivity, and platform access at no charge to KRI for the PoC event only.
- Devices remain Altitude HD property during the PoC unless otherwise agreed in writing; KRI is responsible for reasonable care, return, and loss/theft reporting.
- The PoC is provided “**as is**” for evaluation — no guarantee of uninterrupted service, coverage, or alert delivery.
- Participation does not oblige KRI to purchase hardware or services; Phase 1 terms apply only upon signed acceptance.

3. Supply, ownership & risk

- **Post-PoC purchases:** title to handsets and mounting hardware passes to KRI on payment (or as stated on invoice). KRI bears risk of loss, damage, and theft from delivery onward.
- **Software:** CrewSight application and cloud services remain Altitude HD intellectual property. KRI receives a non-exclusive, non-transferable licence for the agreed season(s) and fleet size.
- **SIMs:** under the data-included option, SIMs are provisioned for KRI’s registered devices; misuse, commercial resale of connectivity, or use outside approved events may breach carrier terms.
- Warranty on handsets is limited to manufacturer terms passed through from supplier; Altitude HD does not warrant hardware beyond supply chain pass-through unless explicitly stated.

4. Limitation of liability

To the maximum extent permitted by New Zealand law, Altitude HD’s liability under any CrewSight agreement is limited to fees paid for the relevant service period (or, for the PoC, a nominal cap to be agreed). Altitude HD is not liable for indirect, consequential, or special loss including loss of profit, reputational harm, or claims arising from reliance on GPS, cellular, or cloud availability. Nothing in this section excludes liability that cannot be excluded under the *Contract and Commercial Law Act 2017* or other applicable law (including the *Consumer Guarantees Act 1993* where it applies to consumer transactions).

5. Privacy & personal data

- GPS tracks and device identifiers may relate to identifiable athletes, coaches, or officials. Collection and use must comply with the **Privacy Act 2020** and any applicable Rowing NZ policies.
- **Event owner** (regatta organiser) is primarily responsible for privacy notices, lawful basis for processing, and retention limits for event data.
- Altitude HD acts as a service provider storing and processing data on documented instructions; a data-processing or service schedule should define roles, subprocessors (e.g. Vercel, Neon, Upstash), retention, and deletion (including purge between regattas where agreed).
- Broadcast overlays and public live streams should avoid displaying personal data beyond what participants have reasonably expected for that event.

6. Data ownership & use

- Raw event data (tracks, alerts, session logs) is owned or controlled by the **event owner**, subject to platform licence terms and privacy law.
- Altitude HD may use aggregated, de-identified operational metrics to improve the service; use of identifiable data for product development or marketing requires separate consent.
- Export and archive formats should be agreed per season; KRI does not acquire rights to national RNZ datasets or third-party event data by virtue of device ownership alone.

7. Third-party & partner hire

- When KRI hires devices to South Island Rowing, Beach Sprints NZ, or other organisers, KRI should use a written **hire agreement** covering custody, damage, return, insurance, and permitted use.
- Hirers must comply with CrewSight acceptable-use terms and event privacy requirements. Altitude HD may suspend ingest for unregistered events or policy breaches.
- Device hire fees are a matter between KRI and the hirer; Altitude HD is not party to hire pricing unless explicitly engaged for event support.

8. Insurance

Parties should confirm adequate cover with their brokers. KRI should consider asset insurance for handset fleet and public liability for device deployment and charging infrastructure. Event organisers typically carry event liability cover; confirm whether hire of tracking equipment affects certificates. Altitude HD maintains standard business insurance appropriate to software and broadcast services; scope to be confirmed in master service terms.

9. Health & safety

Installation of charging racking on the number hut wall must comply with KRI facility rules, electrical safety standards, and any WorkSafe obligations for fixed installations and cable management. Mounting kits must be fitted so as not to impede crew movement or boat equipment. CrewSight does not remove the need for RowSafe-compliant procedures and official patrol coverage.

10. Intellectual property & broadcast

- CrewSight, AHD LiveStream, overlay graphics, and related branding are Altitude HD IP. Licence for on-screen broadcast use should be confirmed with Rowing NZ / Milford where feeds are distributed publicly.
- Future national propositions (analytics, sponsor integrations) remain subject to separate RNZ-AHD commercial agreements; KRI is not granted exclusive national rights through this rollout.

11. Governing law & disputes

Formal agreements are intended to be governed by the laws of **New Zealand**. Disputes should first be addressed through good-faith negotiation between operational contacts; escalation to mediation or courts in New Zealand as specified in executed contracts.

12. Document validity

This plan is valid for **90 days** from the date shown on the cover for commercial indicatives. Legal and compliance positions may be updated when formal terms are issued. Each party should obtain independent legal advice before signing binding documents.